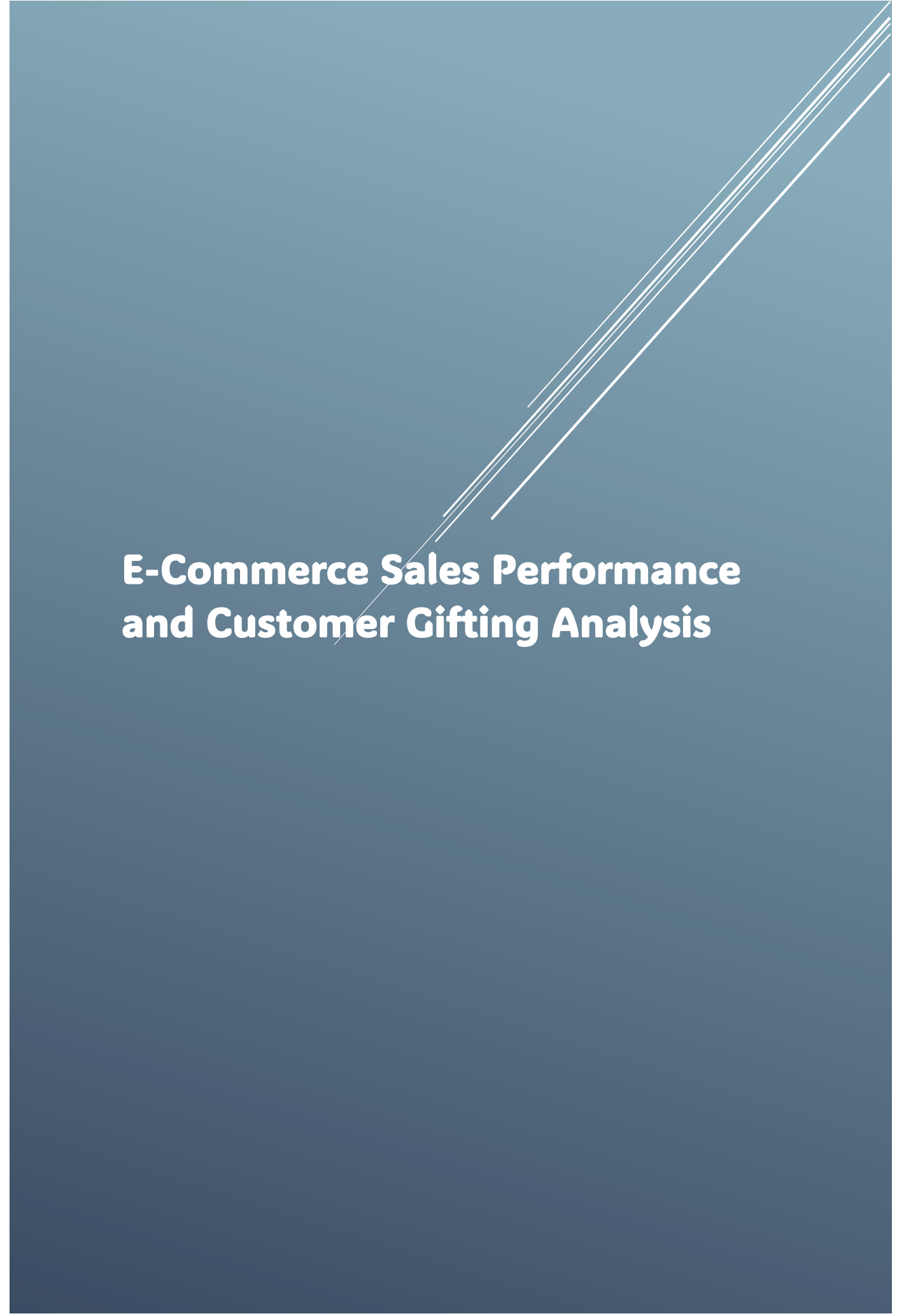


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E-Commerce Sales Performance and Customer Gifting Analysis

1. Project Title

Ferns and Petals (FNP) E-Commerce Sales Performance and Customer Gifting Analysis

2. Project Objective

The primary objective of this project was to analyze transactional, product, and customer datasets from Ferns and Petals (FNP) across 1,000 orders to unlock optimization pathways for their seasonal gifting catalog.

The project aimed to:

- **Establish Relational Integration:** Map separate database entities across orders, unique customers, and product items.
- **Quantify Fulfillment Efficiencies:** Measure operational delivery speeds relative to order volumes to optimize logistic throughput.
- **Decode Gifting Behavior:** Track revenue contributions across target festive occasions (e.g., Diwali, Raksha Bandhan, Valentine's Day) to adjust inventory levels.
- **Build Executive Dashboards:** Design a high-impact corporate report visual to track performance metrics across top cities, seasonal variations, and key product categories.

3. Project Description

Overview

This project presents an end-to-end data processing and analytics architecture analyzing 1,000 consumer purchase histories across 10 distinct operational KPIs. It evaluates order behaviors across geographical hubs and product lines (such as Plants, Cakes, Sweets, and Soft Toys) to maximize e-commerce performance.

Approach & System Engineering

1. Relational Schema Integration (Python & SQL):

- Loaded products.csv, orders.csv, and customers.csv datasets.
- Linked entities together via foundational foreign key fields (Customer_ID, Product_ID).
- Parsed raw dates (Order_Date, Delivery_Date) to dynamically isolate order-to-delivery lifecycle turnarounds.

2. Advanced Core Calculations (DAX & Analytics):

- Engineered data pipelines to extract total revenue metrics by multiplying transaction volumes by item retail prices.
- Filtered geographic distribution vectors to classify leading city performance hubs.

3. Dashboard Design (Power BI):

- Arranged a professional multi-panel grid framework utilizing clean cards, line matrices, bar aggregations, and comparative distribution models.

Technologies Used

- **Data Cleansing & Modeling Frameworks:** Python (pandas, numpy)
- **Relational Query Processing:** PostgreSQL / Structured SQL
- **BI Interface Engineering:** Power BI Desktop (DAX Processing Engine)

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis & understanding FNP operational problem state	Assessed the 10 business questions regarding delivery turnarounds and seasonal trends.
Week 2	Data ingestion, exploration, and constraint validation	Evaluated CSV files using Pandas; handled categorical date formats.
Week 3	Database schema generation and mapping links	Formulated structural keys between customer profiles, order lines, and items.
Week 4	Core logic design and statistical calculations	Wrote analytical queries to measure delivery durations and identify top cities.
Week 5	Inter-season performance calculations	Grouped order trends to map revenue variations by event types.
Week 6	Power BI modeling setup	Connected dataset configurations to the visualization desktop engine.
Week 7	Interactive canvas arrangement and UI formatting	Constructed charts, KPIs, and filtering panels to align with FNP branding themes.
Week 8	Verification, testing, and final packaging	Validated final metrics against base tables and compiled documentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Initial Kickoff	Project alignment, problem definition, and resource environment provisioning	27 Nov 2025
Data Ingestion Signature	Verification of complete cross-table schema links between orders, clients, and items	12 Dec 2025
Analytical Milestone	Processing of targeted business metrics including delivery delays and geographic order counts	28 Dec 2025
Interface Construction	Final visual canvas build highlighting key product lines and seasonal trends	14 Jan 2026
Project Closure	Validation of metrics, documentation sign-off, and final submission pack generation	20 Jan 2026

5b. Project Execution Details

The project successfully answered all 10 core business requirements:

- **Financial Foundations:** Established total gross sales and individual customer spending habits.
- **Supply Chain Tracking:** Calculated the exact average timeline from order confirmation to final customer drop-off.
- **Seasonal Matrix Processing:** Compared financial returns across festive cycles (Valentine's Day, Diwali, Raksha Bandhan) to highlight key popularity points.
- **Geographic Demographics:** Isolated order concentrations to rank the top 10 performing urban centers.

6. Snapshots / Screenshots

(Please embed your dashboard layout and data model screenshot images from your workspace into these sections)

1. **FPN Executive Sales Performance Canvas Dashboard**
2. **Relational Data Schema Model View (Products, Orders, Customers)**
3. **Seasonal Occasion Analysis and Delivery Performance Distribution Matrix**

7. Challenges Faced

1. Multi-Format Date Columns:

- *Challenge:* Mixed text timestamp configurations prevented direct calculations of delivery turnarounds.
- *Resolution:* Standardized all date strings into formal datetime structures prior to computing fulfillment duration metrics.

2. Occasion Attribute Inconsistencies:

- *Challenge:* Occasion variables mapped both at the product level and order log levels threatened category classification logic.
- *Resolution:* Implemented clear hierarchy tracking using the product table as the main definition for core item popularity, and the order table to track situational transaction volume.

3. Geographic Cardinality:

- *Challenge:* Plotting orders across dozens of unique cities over-cluttered the visualization space.
- *Resolution:* Utilized Top-N ranking filters to isolate and display only the top 10 performance centers, keeping visuals clean and scannable.

8. Learnings & Skills Acquired

Technical Skills

- Relational E-Commerce Data Modeling
- Time-Intelligence Formula Configuration
- Geographic Market Performance Stratification
- High-Density Interactive Chart Development

Soft Skills

- E-Commerce Supply Chain Optimization Logic
- Strategic Business Insight Formulation
- Cross-Functional Performance Dashboard Layout Design
- Execution Planning and Structured Milestone Tracking

9. Testimonials from Team

"Working through the Ferns and Petals case study offered fantastic insight into managing complex retail logistics and seasonal data structures. Converting disconnected transaction, customer, and product files into an integrated dashboard highlights how clear data pipelines drive smarter business strategies."

10. Conclusion

The completed solution provides an actionable analytics resource for the FNP ecosystem. By isolating fulfillment velocity trends, ranking top regional market hubs, and showcasing seasonal event revenue patterns, the system offers clear evidence to guide procurement and logistics planning. The model performs efficiently and follows professional design principles.

11. Acknowledgements

I would like to express my sincere gratitude to Infosys for providing me with this valuable internship opportunity. I am deeply thankful to my mentors, Ms. Latha and Ms. Jaya, for their continuous guidance, encouragement, and insightful feedback throughout the internship. I would also like to extend my appreciation to my team members for their collaboration, support, and teamwork, which played a significant role in the successful completion of this project.